



A central instrument in the government's power sector reform toolkit is the prepaid smart meter.

Smart Metering Impact

Immediate Revenue Gains

- 100% arrears recovery (select divisions)
- Collection efficiency: 100%+ improvement
- Revenue per consumer: +65%
- Reduced billing disputes



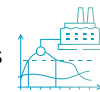
Bihar, a Classic Case Study

- Historically loss-making discoms
- Now profitable (FY2024)
- Smart meters key contributor
- Improved financial discipline



Liquidity Cascade Effect

- Faster generator payments
- Reduced receivable delays
- Stronger transmission cash flows
- Stabilised power value chain



Regulatory Discipline Layer

- CERC tariff oversight
- State commission monitoring
- RDSS reform compliance linked
- Capital modelling clarity improves



Social Execution Strategy

- Local workforce deployment
- Consumer awareness drives
- Reduced resistance rates
- Lower revisit costs



Margin Dynamics in Smart Metering

Smart metering is competitive. Bids are won through tariff commitments over long periods and margins are shaped by procurement discipline, installation efficiency, communication uptime and maintenance responsiveness.

AESL partners with tier-1 original equipment manufacturers to reduce quality variability and supply risk. It also holds strategic interests in AdaniConnex, a data centre joint venture and Esyasoft, a smart grid and digital energy solutions company. These relationships support integration between hardware and data systems.

Vertical integration can enhance system reliability and compress response times when faults occur. In performance-linked contracts, uptime directly influences revenue realisation.

The Regulatory Overlay

India's power sector operates within layered regulatory frameworks. Transmission tariffs are approved by the CERC for interstate assets while state electricity regulatory commissions oversee distribution performance and consumer tariffs.

Under the RDSS, funding disbursement is linked to reform compliance. States must demonstrate improvements in billing efficiency, reduction in AT&C losses and infrastructure upgrades. Smart metering thus sits at the intersection of policy compliance and commercial viability. Delays in installation can affect both contractor cash flows and state eligibility for central support.

For private players, regulatory clarity reduces uncertainty. Stable tariff frameworks and defined performance metrics allow capital providers to model returns more confidently.

Operational Outcomes and Financial Impact

Adoption to smart meters provides nearly instant results. In divisions where smart meter penetration is high, AESL reports 100 per cent arrear recovery and collection efficiency improvements exceeding 100 per cent compared with legacy systems. Revenue per consumer has increased by up to 65 per cent in those divisions, according to internal company data. Bihar, where discoms have never made a profit till 2024, now hosts profitable state-owned discoms and smart meters are a key contributor to this financial turnaround.

While these improvements are specific to certain geographies and not yet representative of national averages, they highlight the economic logic of smart metering. Accurate billing reduces disputes. Prepaid functionality shortens cash cycles and remote disconnection limits accumulation of unpaid dues. Improved discom liquidity has second-order effects.

Generators receive timely payments while transmission companies experience reduced receivable delays. In a nutshell, the broader power value chain stabilises.

Socio-operational Realities

Execution at national scale requires social alignment. India's linguistic and cultural diversity presents operational complexity. AESL recruits installation teams locally to leverage familiarity with regional conditions. Consumer awareness campaigns precede meter installation to explain benefits. This reduces friction during installation.

Resistance reduction is not merely reputational. High revisit rates increase costs and delay rollout milestones. In a programme targeting hundreds of millions of installations, social friction